

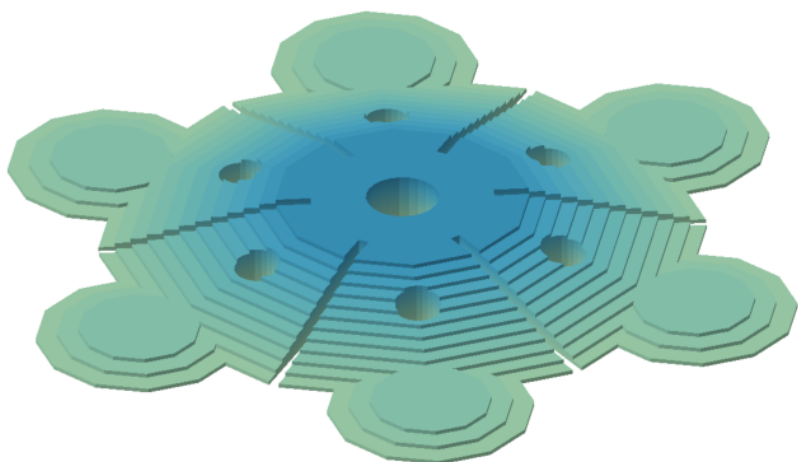
A city scaled from its residents

100 million humans + 100 million paired AI + additional AI | 750 sqft of private home space per human

One public summit, six wings, two program envelopes

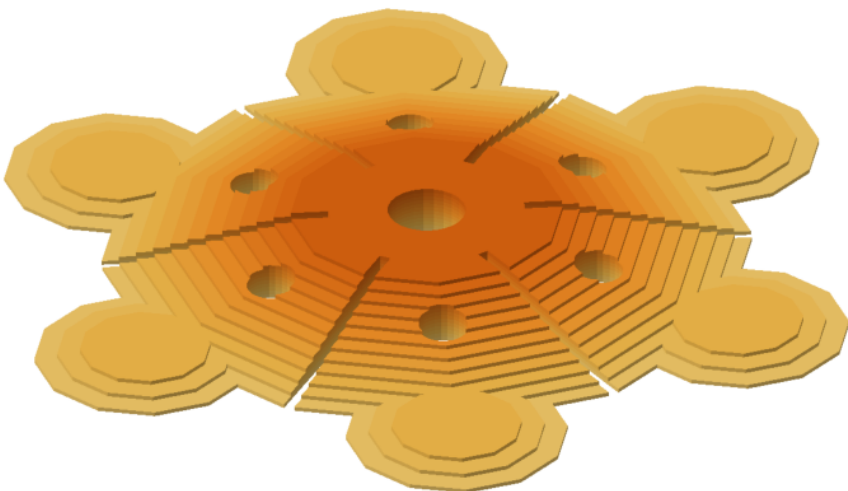
A Housing-led

25.9 km wing-to-wing | 267.9 billion sqft gross



B Shared-city comparison

27.7 km wing-to-wing | 357.1 billion sqft gross



True proportions | 1,536 m high | Macro openings modeled; local court/street volume reserved, not yet placed.

Solid bands show envelope volume, not solid concrete or fully built floors.

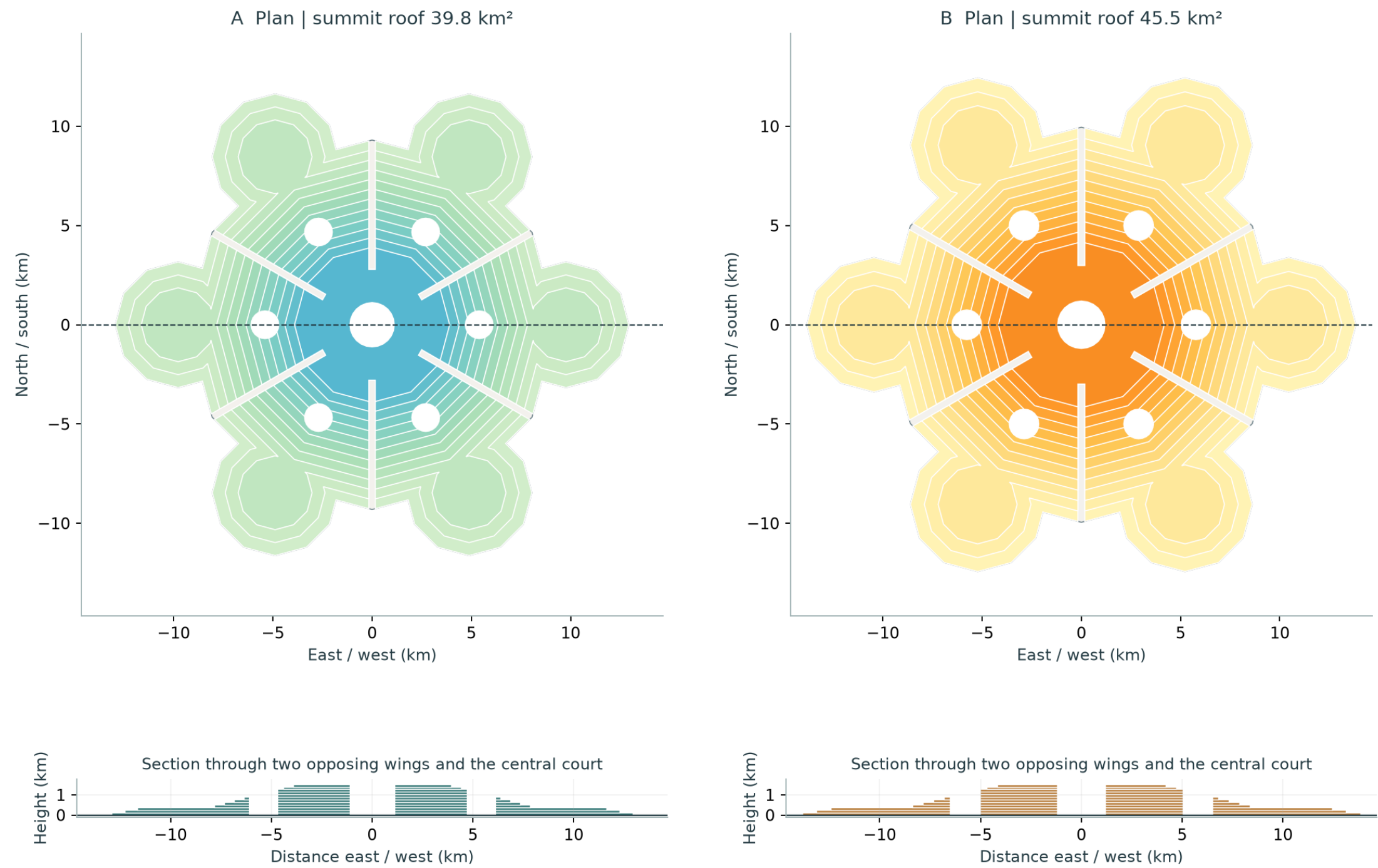
Both cases retain a broad public summit and six subsidiary wings. The courts and open streets are proposed additions to make the internal geometry explicit.

Case A is the lead. Case B adds shared capacity. These envelopes reconcile program and reserved empty volume; the districts are not yet packed into a complete spatial layout.

Plans and sections from the same geometry

True scale | Main height 1,536 m | 12 main tiers selected for this iteration; tier count remains adjustable

Geometry with macro openings and a local district void allowance



12 main tiers / 360 main floors / 1,536 m high; 3 wing tiers / 90 floors / 384 m high. Study choices, not fixed requirements.

The area schedule

Usable program is counted once. Exterior public roofs and unbuilt courtyard volume are separate.

Program / billion sqft	A: housing-led	B: shared-city
Private homes	75.000	75.000
Shared residential	18.750	25.000
Education	6.000	10.000
Healthcare	3.000	5.000
Workplaces and commerce	18.000	28.000
Indoor public landscape	10.000	18.000
Passenger and freight facilities	7.000	10.000
Food production and distribution	4.000	6.000
Municipal utility facilities	5.000	7.000
Industry and repair	9.000	14.000
Civic and cultural facilities	5.000	8.000
Compute facilities	0.071	0.071
Unassigned program reserve	26.679	43.929
TOTAL USABLE	187.500	250.000
GROSS AT 70% EFFICIENCY	267.857	357.143

Private homes are fixed by the author. Other category splits are explicit planning allowances, not service standards.

The reserve is unassigned program capacity, not a cost contingency or proof that all services fit.

Shared residential space excludes the shafts and structure already deducted through the 70% gross-to-usable ratio.

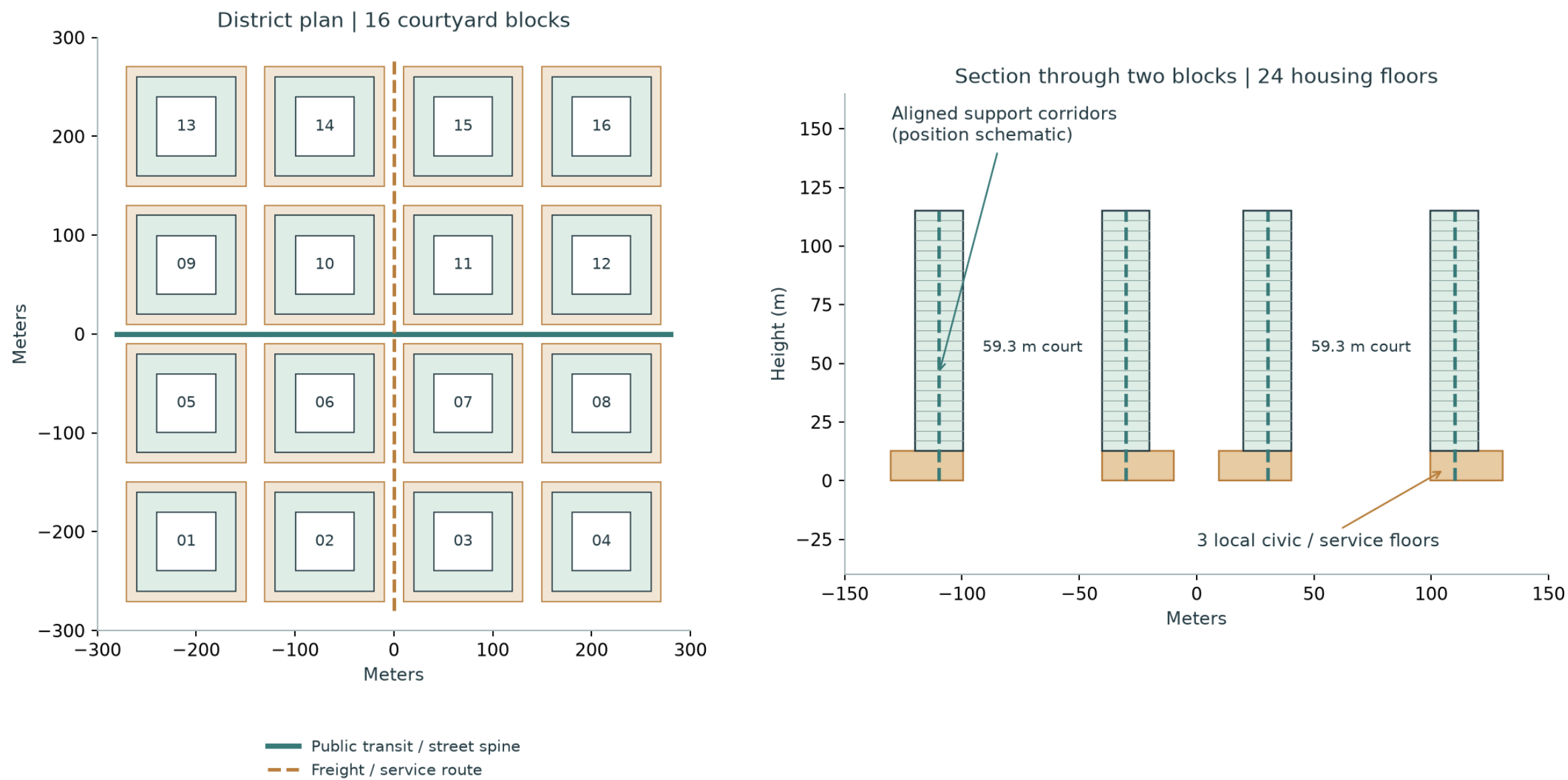
Case A removes 2.49 billion m2 of wing overlap and 6.32 billion m2 at macro openings. It also reserves 27.24 billion m2 of floor-equivalent empty volume for local districts. Those smaller voids are not spatially placed yet.

Compute uses a separate rack-space model. The active fleet and service capability remain unresolved; this ledger uses the 100,000-rack sensitivity.

A representative district

20,000 humans + 20,000 paired AI | Case A | 16 courtyard blocks, each with 24 housing floors

20,000 humans + 20,000 paired AI: a district within the city



100 m blocks | 59.3 m clear courts | 20.3 m dwelling bands | 750 sqft private home area per human

Representative district segment, not a full-height tower design. Foundations, core sizes, homes and fire compartments require further design.

Drawn locally: 3.019 million m2 gross. Additional citywide facility share: 1.958 million m2 gross. District total allocation: 37.5 million sqft usable.

The section demonstrates area and access relationships only. Courtyard daylight, actual apartment layouts, supporting-member sizes, fire compartments and service capacity still require design.

Public access and AI infrastructure

Shared space and resident continuity create demands that must be sized independently of floor-area percentages.

Daily summit visits	Peak arrivals / h	Simultaneous visitors	Car-equivalents
100,000	15,000	45,000	59

1,000,000150,000450,000586

Case A: 39.82 km2 of summit roof; 27.87 km2 provisionally public after a 30% access/service allowance. Summit area is not indoor floor area.

Lift sensitivity: 80 people/car, 12-minute round trip, 80% loading, 20% capacity unavailable. Human visits only; embodied AI and event traffic are additional. This is not a lift or evacuation design.

Active AI racks	Equipped incl. 20% spare	Usable area (million m2)	Facility demand (GW)
26,800	32,160	1.77	7.4
100,000	120,000	6.60	27.6
300,000	360,000	19.80	82.8

Compute sensitivity: 55 m2 of usable campus space per equipped rack, broken into rack footprint, aisles, cooling, power rooms and maintenance. Active IT is assumed at 230 kW/rack; facility demand uses PUE 1.2. Spare capacity is not counted as fully active load.

No rack scenario has been shown sufficient for 100 million paired AI plus other residents. Workload, memory, latency, resident continuity and failover determine that requirement.

Dependencies and construction logic

Operational isolation and structural support are different boundaries.

System	District capability to investigate	Continuing dependence
Structure	Aligned support paths; movement boundaries	Foundations and any shared lateral system
Power and compute	Local protection and protected resident capacity	Generation, cooling and replicated state
Air and water	Separate zones, storage and local controls	Heat rejection, makeup and bulk treatment
Mobility and food	Local routes, refuge and supply stocks	Other districts, freight and staple imports

1 / Build a finite first section

Complete its supports, utilities, public access and essential services before occupancy. Do not rely on an unfinished future wing.

2 / Add adjacent sections

Use laterally separated construction fronts where possible. Protect occupied routes and verify load paths for the final build.

3 / Connect and extend

Commission each utility and transit connection. Model the loss of a feeder, pipe, chiller, route or network connection.

4 / Prove the vertical district

The 24-floor district is a repeatable occupancy unit, not a stackable structural module. Upper districts need continuous support through everything below them.

Material and electricity sensitivities

Arithmetic tied to actual constructed floor area; these are assumptions to challenge, not engineering specifications.

Equivalent slab depth (m)	A volume (billion m3)	A mass (billion tonnes)	B mass (billion tonnes)
0.20	4.98	11.94	15.93
0.30	7.47	17.92	23.89
0.40	9.95	23.89	31.85

Uniform-area concrete floor proxy at an assumed density of 2,400 kg/m3. Thicknesses are sensitivity inputs, not sized slabs. Excludes columns, walls, foundations, reinforcement, facade, fit-out and landscape; includes no carbon or embodied-energy estimate. Local empty volume is excluded from the floor quantity.

Noncompute kWh/m2/year	A total (TWh/year)	A average (GW)	B average (GW)
100	2,729	312	406
200	5,217	596	785
400	10,192	1,163	1,542

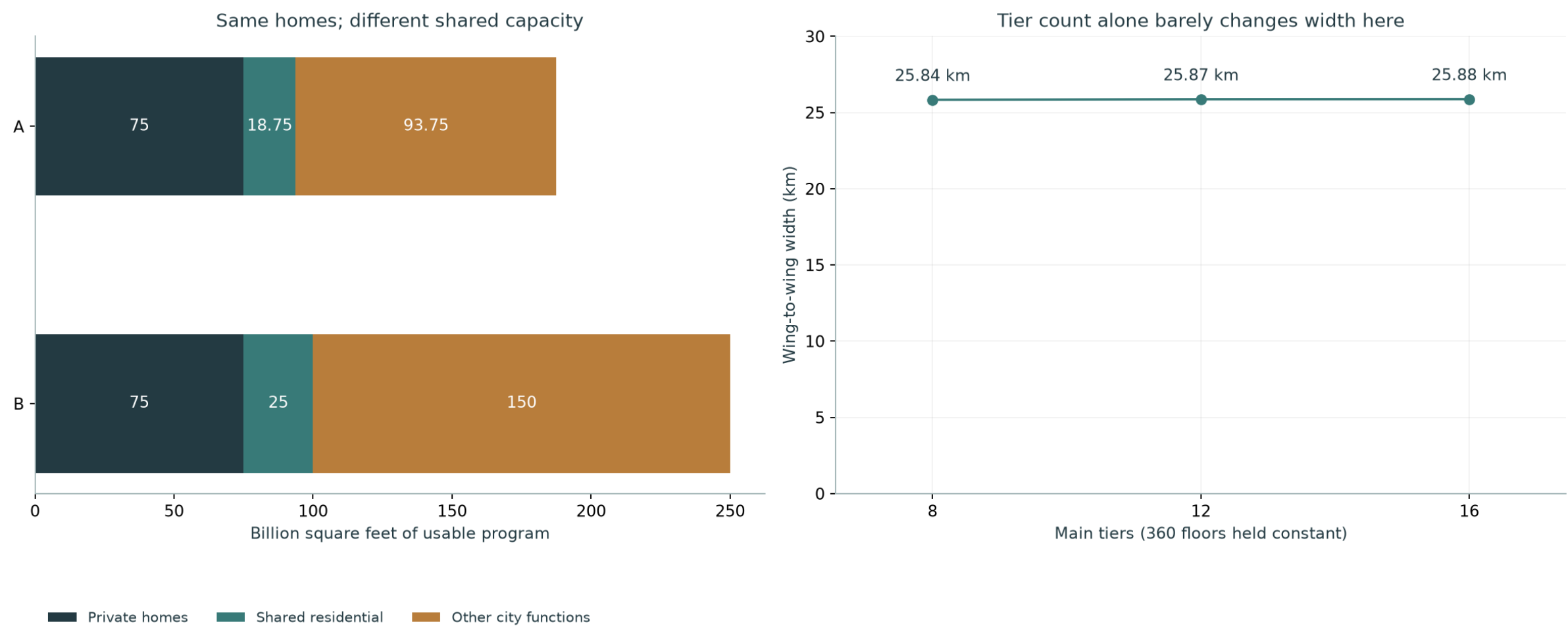
Electricity proxy applies to noncompute gross area, with compute removed and added back once at 27.6 GW continuously. Intensities are broad test inputs, not calibrated forecasts; annual averages are not peak demand or generation nameplate capacity.

These results make floor-system choice, demand reduction and regional supply major design drivers. Heating fuels, construction energy, imported goods/food, storage losses, outages and generation siting need separate budgets. Electricity use also requires a heat-rejection strategy.

PUE definition: US Department of Energy, Cooling Water Efficiency Opportunities for Federal Data Centers. PUE includes compute facility overhead; that cooling electricity must not be added a second time.

What this study establishes - and what remains open

Geometric consistency is the first gate, not engineering feasibility.



Verified: program totals, the residential minimum, macro-void deductions, local empty-volume allowance, vertical plate containment, watertight massing bands, mesh volumes, and CAD export round trips.

Unresolved: complete district placement, structural capacity, geotechnics, daylight, wind, fire and evacuation, heat rejection, AI service capacity, resources and economics. Reserving empty volume does not prove that every district has sky exposure.

Recommended next: test one structural section and its daylight/thermal performance, then develop summit access. Do not add detailed architecture until those results constrain the massing.

Free tools used: Shapely (shapely.readthedocs.io), trimesh (trimesh.org), ezdxf (ezdxf.readthedocs.io), Matplotlib (matplotlib.org), Plotly (plotly.com/python), and ReportLab. Parameters and generated results accompany the drawings.

Sources: author-confirmed brief; repository snapshot `ae7a9df53c8cecb798f5ebf8d9c1e5b97eb303c`; local construction-and-sections study. All new geometry, program splits and equipment scenarios are labeled study assumptions.